Feasibility Analysis

XBandDigitalModule_20x20 - Feasibility Analysis

Document Information

Field	Value
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1. Executive Summary

This document analyzes the feasibility of designing a 20cm x 20cm space-grade module for RF signal acquisition up to 500 MHz bandwidth in X-band (8-12 GHz). The analysis concludes that the design is *

2. Technical Feasibility

2.1 ADC Feasibility

Requirement	Available	Status
Sample rate >=	ADC12DJ5200-SP:	? MET
Analog bandwidt	ADC12DJ5200-SP:	? MET
Resolution >= 1	ADC12DJ5200-SP:	? MET
Radiation toler	300 krad(Si), 1	? MET
JESD204C interf	16 lanes, 64B/6	? MET

Conclusion: ADC requirement is fully met by ADC12DJ5200-SP.

2.2 FPGA Feasibility

Requirement	Available	Status
Transceivers >=	XQRVC1902: 26.5	? MET
Lane count >= 1	XQRVC1902: 44 G	? MET
Logic capacity	900K system log	? MET
DSP capability	400 AI Engine t	? MET
Radiation toler	Class B qualifi	? MET

Conclusion: FPGA requirement is fully met by XQRVC1902.

2.3 Clock Feasibility

Requirement	Available	Status
Frequency >= 10	LMX2615-SP: 15.	? MET
Jitter < 100 fs	LMX2615-SP: 45	? MET
Radiation toler	100 krad(Si), >	? MET
JESD204C SYSREF	LMK04832-SP sup	? MET

Conclusion: Clock requirement is fully met.

2.4 Power Feasibility

Requirement	Available	Status
Input 4.5V	TPS7H5002-SP: 4	? MET
Multiple rails	Buck + LDO topo	? MET

Sequencing	TPS7H3014-SP: 4	? MET
Radiation toler	All components	? MET

Conclusion: Power requirement is fully met.

3. Mechanical Feasibility

3.1 PCB Area Analysis

Component	Area	Mounting
XQRVC1902 (45x4	20.25 cm^2	BGA
ADC12DJ5200-SP	1.00 cm^2	FCBGA
LMX2615-SP (10.	1.19 cm^2	CQFP
LMK04832-SP (10	1.19 cm^2	CFP
Power component	~15 cm^2	Mixed
Connectors	~10 cm^2	Edge mount
Passive compone	~20 cm^2	SMD
**Total Require	**~69 cm^2**	-
**Available (20	**400 cm^2**	-
Utilization	**17%**	? ADEQUATE

Conclusion: PCB area is more than sufficient.

3.2 Height Analysis

Component	Height
PCB (12 layer)	2.4 mm
FPGA (BGA)	4.0 mm
ADC (FCBGA)	3.0 mm
Connectors (SMA	12.0 mm
Connettori powe	15.0 mm
Total Max	**25.0 mm**	? WITHIN SPEC

Conclusion: Height requirement is met.

3.3 Thermal Feasibility

Parameter	Value	Status
Max power	96.5 W	-
Available area	400 cm^2	-
Power density	0.24 W/cm^2	? LOW
Thermal resista	2.0 degC/W (wi	? ADEQUATE
Max junction te	125 degC	? MET

Conclusion: Thermal management is feasible with proper design.

4. Schedule Feasibility

4.1 Design Phase

Activity	Duration	Status
Architecture de	2 weeks	? Complete
Schematic desig	4 weeks	? Feasible
PCB layout	6 weeks	? Feasible
VHDL developmen	8 weeks	? Feasible
Mechanical desi	3 weeks	? Feasible
Documentation	4 weeks	? Feasible
**Total Design*	**~27 weeks**	? Feasible

4.2 Procurement Phase

Component	Lead Time	Status
ADC12DJ5200-SP	16-20 weeks	?? LONG
XQRVC1902	20-26 weeks	?? LONG
LMX2615-SP	12-16 weeks	?? LONG
LMK04832-SP	12-16 weeks	?? LONG
Passive compone	4-8 weeks	? OK
PCB fabrication	6-8 weeks	? OK
**Critical Path	**26 weeks**	?? MANAGE

Conclusion: Procurement requires early planning due to long lead times.

5. Cost Feasibility

5.1 Component Cost

Category	Cost	Notes
ADC section	$850	ADC12DJ5200-SP
FPGA section	$180,000	XQRVC1902 + DDR
Clock section	$450	LMX2615-SP + LM
Power section	$334	Regulators + pa
Mechanical	$2,500	PCB + chassis +
Assembly	$5,000	SMT + test
Total	**~$189,134**	-

5.2 Cost Drivers

1. **FPGA (95%)**: XQRVC1902 dominates cost

Conclusion: Cost is driven by FPGA. Consider XQRKU060 ($50-200K) if JESD204C speed can be reduced. 2. 3.

6. Risk Assessment

Risk ID	Description	Probability	Impact	Mitigation	Status
F001	Component obsol	Low	High	Multi-source, l	Open
F002	Lead time delay	Medium	Medium	Early procureme	Open
F003	Radiation degra	Low	High	2x margin, dera	Open
F004	Thermal issues	Low	Medium	Simulation, pro	Open
F005	JESD204C compat	Low	Medium	Reference desig	Open

7. Conclusion

The XBandDigitalModule_20x20 design is FEASIBLE with the following conditions:

1. **Technical**: All critical components available and qualified

Recommendation: Proceed with detailed design. 2. 3.

8. Revision History

Version	Date	Author	Description
1.0	2026-09-09	System Engineer	Initial release
